



TED UNIVERSITY

SENG 491

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SkillHood

Project Specifications Report

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1. Introduction

1.1 Purpose

SkillHood aims to provide users with the ability to share their skills and services in a sustainable manner while eliminating the requirement of financial transaction. This document will outline the core requirements of the system which will focus on developing a collaborative environment through a credit-based system.

1.2 Product Scope

SkillHood is a software application that enables users to exchange skills and services using a unique internal credit system. The platform uses a value-based credit system instead of a conventional payment approach, allowing users to earn credits for teaching and use those credits to learn from other members. This project's primary goal is to create a community of people who are free from financial constraints, have access to each other's expertise, and can help one another.

This project's scope consists of creating an application that will support the two forms of exchange: Live Interactive Services (tutoring, mentoring, etc.) and Service-Based Exchanges (gardening, dog walking, etc.). Credits will be given by the application based on the user's reputation, demand, and skill level. In order to build trust, it will also have user profile pages, service listing sites, scheduling tools, credit tracking tools, and feedback mechanisms.

2. Overall Description

2.1 Product Perspective

SkillHood goes far beyond being an application. It is a continuously evolving social community for everyone to share their knowledge and skills to teach and learn. Every time a member of SkillHood provides some type of service (mentorship, tutoring, etc.), they will always create some type of value that they are able to use to learn something new. Therefore, SkillHood creates a continuous cycle of learning, engagement, and mutual benefit. SkillHood also presents itself as an innovative solution to the sharing economy, where the value is

determined by how individuals can assist one another instead of by how much money they have.

2.2 Product Functions

The major functions the product will perform include:

- Skill Marketplace
 - Allows users to view the available skills, using filters and categories to find the appropriate skills.
- Scheduling System
 - Provide booking options for each session using a calendar or time slot format.
- Matching Algorithm
 - Users are connected to each other using common skill set and performance metric.
- Value-Based Credit System
 - Track (and provide) both the earned and spent credit points, depending on the skill level, how many are required/available for the requested session and the user's past performance.
- Communication and Notifications
 - Provides users with real-time updates regarding interactions with others (session and system) to encourage participation.
- Reputation and Review System
 - Allows users to provide ratings to each other after they have completed sessions to help foster trust and assure the quality of services provided.

2.3 Operating Environment

The mobile application frontend shall be built using React Native, the backend shall be implemented with Node.js and Express, and the database shall be managed using PostgreSQL.

2.4 Constraints

Several constraints will influence the development and implementation of this project:

- **Transaction Integrity Constraint:** The transfer of credit points should only occur once the activity has been completed and verified by both users, to protect against fraudulent or manipulated transactions.
- **Scheduling Constraint:** Users will be able to view their own schedules, and the system will prevent scheduling conflicts, so no user will be scheduled to participate in conflicting sessions.
- **Scalability Constraint:** The system shall maintain responsiveness and performance as the number of users increases.
- **Economic Constraint:** The platform cannot facilitate monetary exchange; therefore, it must minimize operating costs and system infrastructure expenses.
- **Social Constraint:** The system will have to accommodate users of different backgrounds and cultural identity. The system must be able to avoid discrimination and allow equal opportunities for all users.
- **Legal Constraint:** The platform must obey laws and regulations for data protection and privacy.

2.5 Professional and Ethical Issues

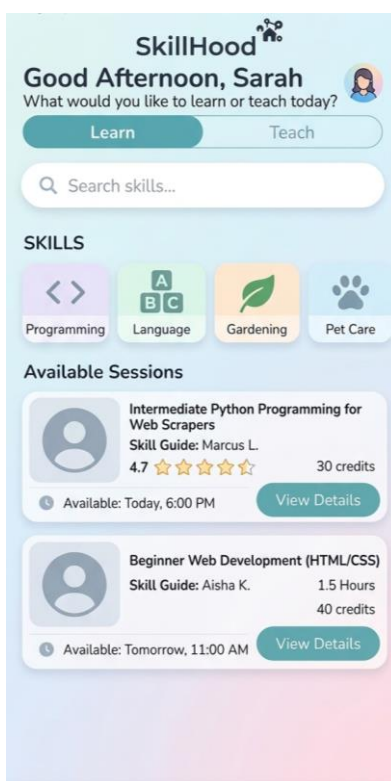
- **User Privacy:** Protecting users' personal information by committing to data protection rules. Any information provided by users is required to be processed and transferred in a way that preserves its confidentiality and integrity.
- **Fairness and Transparency:** Controlling the platform's credit system in order to assure transparency while preventing abuse or manipulation.

- **Trust and Safety:** Creating a safe and professional environment by allowing users to report inappropriate behavior and taking action to resolve violations.
- **Data Integrity:** Recording reliable records of all system activity (including credit transactions) and avoiding unauthorized modification or exploitation of system data.
- **Inclusivity:** Designing the platform to include users from all backgrounds, skill levels, and skills, while avoiding discriminatory behaviors in all system operations.

3. External Interface Requirements

3.1 User Interfaces

The designed interface for SkillHood app enables both learning (live exchanges) and practical support (service-based exchanges) in an accessible way.



Picture 1. User Interface Mock-up (generated using AI-assisted design tools and refined by the authors)

Interface Characteristics

- Access through a mobile application
- Clean and minimal layout for small screens
- Key components:

- Home screen
- Profile page (skills, ratings, credits)
- Skill discovery and filtering
- Create/manage exchanges
- Scheduling/view sessions
- View credit and transactions
- Live Interactive Exchanges
- Service-Based Exchanges

3.2 Hardware Interfaces

The platform can operate on a variety of consumer devices without any need for specialized hardware. Users will be able to access via their smartphones and tablets all day. Standard internet connectivity will be enough. No extra dependencies like sensors are required.

3.3 Software Interfaces

Management of data, communication services, and security come forward for SkillHood app.

Interface Characteristics

- Backend database:
 - User profiles
 - Records of exchanges
 - Skill listings
 - Transactions
- Authentication mechanisms (security)
- Integration with:
 - Notifications (email/in-app notifications)
 - Scheduling with calendar operations
- App communication through standard network protocols
- Compatibility with modern mobile operating systems (Android/iOS)

4. System Features

4.1 User Account Management

Feature	Description
Actors	User
Stimulus	Enter registration details, profile information, skill entries
Response	User profile, updated account data
Functional Behaviour	Users can register, log in, edit profiles, list skills (offered/requested), specify skill level and exchange type, and view credit balance, history, and ratings.

Table 1. User Account Management: Allows users to create/manage their personal profiles and other data.

4.1.1 Functional Requirements

- **REQ-01.01:** The system shall allow users to register with requested personal information.
- **REQ-01.02:** The system shall allow login to users via their valid credentials.
- **REQ-01.03:** The system shall allow users to edit and update their profile information.
- **REQ-01.04:** The system shall permit users to add/update/remove offered skills.
- **REQ-01.05:** The system shall enable users to select exchange type and skill level.
- **REQ-01.06:** The system shall display the user's credit, previous exchanges and ratings.

4.2 Skill Listing and Discovery

Feature	Description
Actors	User
Stimulus	Open skill listings, filter (category, type, rating, availability)
Response	Filtered skill listings loads
Functional Behaviour	Users can create listings for Live or Service-Based exchanges, include descriptions and duration, and search/filter listings based on multiple criteria.

Table 2. Skill Listing and Discovery: Enables users to publish/search for skills.

4.2.1 Functional Requirements

- **REQ-02.01:** The system shall offer users to create list of skills.
- **REQ-02.02:** The system shall request each listing to have an exchange type, category and description.
- **REQ-02.03:** The system shall enable users to filter skills using keywords.
- **REQ-02.04:** Based on search criteria, the system shall present matching skill listings.

4.3 Dual Exchange System

4.3.1 Live Interactive Exchange

Feature	Description
Actors	Teacher, Learner
Stimulus	Input schedule details of a live exchange(date, time, duration)
Response	Scheduled sessions, confirmed exchanges loads
Functional Behaviour	Users schedule sessions, system checks availability, both participants attend, and both must confirm completion for credit transfer.

Table 3. Live Interactive Exchange: Supports real-time learning sessions between users.

4.3.1.1 Functional Requirements

- **REQ-03.01.01:** The system shall support users for scheduling live sessions with selecting duration and date of the session.
- **REQ-03.01.02:** Before confirming a session, the system shall validate if the related users are available.
- **REQ-03.01.03:** The system shall send a notification to both sides of users after scheduled sessions.
- **REQ-03.01.04:** The system shall request from both users to confirm a session completion.
- **REQ-03.01.05:** Only after mutual confirmation, the system shall transfer credits.

4.3.2 Service-Based Exchange (Practical Support Exchange)

Feature	Description
Actors	Service Provider, Requester
Stimulus	Input description of task, duration, optional deadline
Responses	Completed service, confirmed exchange output
Functional Behaviour	Users post requests, others accept. Task progresses through states. Requester confirms completion and credits are transferred.

Table 4. Service Based Exchange: Supports asynchronous, outcome-based services without real-time interaction (Only the service provider is there at that time).

4.3.2.1 Functional Requirements

- **REQ-03.02.01:** The system shall enable users to create requests for services with duration and descriptions.
- **REQ-03.02.02:** The system shall allow users to accept requests for services.
- **REQ-03.02.03:** The system shall manage status per service (Requested/Accepted/In Progress/Completed).
- **REQ-03.02.04:** The system shall provide the requester to confirm completion of the service.
- **REQ-03.02.05:** The system shall handle credit transfer after confirmation.

4.4 Value-Based Credit System

Feature	Description
Actors:	System
Stimulus	Input base rate, skill category, user rating, demand data
Response	Load credit value
Functional Behaviour	Credits are computed using a formula combining base rate, skill multiplier, demand factor, and reputation factor. The system displays all components before confirmation.

Table 5. Value-Based Credit System: Calculates credits (dynamically) based on skill value, demand, and reputation.

4.4.1 Functional Requirements

- **REQ-04.01:** The system shall validate credits for services using the following formula:

$$\text{BaseRate} \times \text{SkillValue} \times \text{DemandFactor} \times \text{ReputationFactor}.$$
- **REQ-04.02:** The system shall provide a base rate that is configurable for calculating credit value.
- **REQ-04.03:** The system shall assign a SkillValue multiplier based on the category or tier of the skill.
- **REQ-04.04:** The system shall calculate a DemandFactor with marketplace data.
- **REQ-04.05:** Based on user ratings, the system shall make a calculation for ReputationFactor.

- **REQ-04.06:** Before confirmation, the system shall present all components of the calculation.

4.5 Reputation and Rating System

Feature	Description
Actors:	Users
Stimulus	Ratings, feedback after exchanges as input
Response	Calculate average rating, reputation score
Functional Behaviour	Users rate each other after exchanges. Ratings affect reputation scores and influence matching and credit calculations.

Table 6. Reputation and Rating System: Maintains trust and quality through user feedback.

4.5.1 Functional Requirements

- **REQ-05.01:** The system shall enable users for rating one and other after exchange completion.
- **REQ-05.02:** The system shall allow users to input extra feedback when the exchange is completed.
- **REQ-05.03:** The system shall upload an average rating score for each user.
- **REQ-05.04:** The system shall determine the reputation factor using rating scores per user.

4.6 Matching and Recommendation System

Feature	Description
Actors	System
Stimulus	Skill data, availability, ratings are listed
Response	Update suggested matches, ranked listings
Functional Behaviour	The system matches users based on skill compatibility, availability, exchange type, and reputation, and presents ranked results.

Table 7. Matching and Recommendation System: Suggests compatible users for exchanges.

4.6.1 Functional Requirements

- **REQ-06.01:** Based on compatibility of their skills, the system shall match users.

- **REQ-06.02:** The system shall ensure availability before generating matches.
- **REQ-06.03:** For ranking the matches, the system shall consider user ratings.
- **REQ-06.04:** The system shall recommend matches to users.

4.7 Scheduling and Availability Management

Feature	Description
Actors	User
Stimulus	Availability slots, scheduling requests searched
Response	Scheduled sessions, updated calendar loading
Functional Behaviour	Users define availability, system prevents conflicts, allows session scheduling, modification, and cancellation, and tracks session states.

Table 8. Scheduling and Availability Management: Manages user availability and session scheduling.

4.7.1 Functional Requirements

- **REQ-07.01:** The system shall allow users to define available time slots.
- **REQ-07.02:** The system shall restrict conflicts when scheduling.
- **REQ-07.03:** The system shall permit users to display future sessions.
- **REQ-07.04:** The system shall enable users to update/cancel sessions that are scheduled.

4.8 Credit Balance and Transaction Management

Feature	Description
Actors	System, User
Stimulus	Completed exchanges, credit operations are given
Response	Update balances, transaction history
Functional Behaviour	The system maintains credit balances, logs transactions, ensures credits transfer only after confirmation, and prevents duplicate transactions.

Table 9. Credit Balance and Transaction Management: Tracks and manages all credit-related transactions.

4.8.1 Functional Requirements

- **REQ-08.01:** The system shall record a credit balance per user.
- **REQ-08.02:** The system shall store all transactions.

- **REQ-08.03:** The system shall provide their transaction history to users.
- **REQ-08.04:** The system shall track and restrict credit transfer duplication.

4.9 Economic Stability Mechanisms

Feature	Description
Actors	System
Stimulus	Demand data, skill distribution inputted
Responses	Show adjusted multipliers
Functional Behaviour	The system applies multiplier caps/floors, smooths demand fluctuations, and adjusts values to prevent inflation and imbalance.

Table 10. Economic Stability Mechanisms: Maintains balance in the platform's credit economy.

4.9.1 Functional Requirements

- **REQ-09.01:** For maximum multiplier, the system shall enforce a cap.
- **REQ-09.02:** The system shall ensure a minimum multiplier floor.
- **REQ-09.03:** The system shall handle values based on supply/demand percentage.

4.10 Exchange Confirmation and Validation

Feature	Description
Actors	Users, System
Stimulus	Completion of confirmations
Response	Load approved or pending exchanges
Functional Behaviour	The system applies multiplier caps/floors, smooths demand fluctuations, and adjusts values to prevent inflation and imbalance.

Table 11. Exchange Confirmation and Validation: Ensures exchanges are verified before credit transfer.

4.10.1 Functional Requirements

- **REQ-10.01:** Before credit transfer, the system shall request confirmation.
- **REQ-10.02:** The system shall ask both sides to confirm before live and service-based exchanges.
- **REQ-10.03:** Unless confirmation is completed, the system shall prevent credit transfer.

5. Nonfunctional Requirements

5.1 Performance Requirements

- **Response Time:** The system shall respond to user interactions such as page loading and filtering in at most 2 seconds.
- **Concurrency:** The database and backend architecture shall support up to 500 active users without any performance loss.
- **Credit Calculation Speed:** Dynamic credit factors and complex equations shall be calculated in less than 100 ms.

5.2 Safety Requirements

- **Preventing Data Loss:** The system shall apply ACID principles in case of dysconnectivity during credit calculations.
- **Error Handling:** The system shall forward users to a safe logout page and suspend their incomplete credit transfers during a crash.

5.3 Security Requirements

- **Encryption:** User passwords shall be stored being hashed by bcrypt algorithm. All data transactions have to be done over TLS 1.3 and HTTPS protocols.
- **Authorization:** JSON Web Token based authentication shall be used for API access.

5.4 Software Quality Attributes

- **Usability:** The interface shall be usable by users with vision problems such as color blindness by designing the interface based on WCAG 2.1 standards.
- **Reliability:** System's uptime shall be at least 99.5% on a yearly basis.
- **Maintainability:** The code structure of the system shall be modular and shall have extensive documentation.

6. Other Requirements

6.1 Database Requirements

- **Relational Structure:** The complex relationship between user reputation, expertise level, and credit history will be managed by PostgreSQL.
- **Backups:** The database shall be backed up every 24 hours automatically, and critical transaction logs shall be stored in a separate and secure storage.
- **Data Integrity:** The system shall implement Foreign Key constraints, in order to ensure that incomplete credit transactions are not lost in the event of an account deletion.

6.2 Legal and Regulatory Requirements

- **KVKK/GDPR Compliance:** The system shall obtain consent from users for data storing and processing. The system shall also support the “right to be forgotten”, allowing users to delete all of their personal data from the system, in accordance with regulations.
- **Financial Regulations:** The platform shall comply with the local non monetary trade regulations.

6.3 Internationalization and Localization

- **Language Support:** Initially, the system shall support Turkish and English languages. All static texts shall be stored in language files (i18n).
- **Time Zones:** Scheduling tools shall function dynamically according to the users’ local time zones.

6.4 Reusability and Extensibility

- **API Scalability:** In order to integrate the application to institutional education platforms in the future, the RESTful API shall be developed to conform with OpenAPI/Swagger standards.
- **Modular Components:** The UI components on the React Native side shall be configured in such a way that they can be reusable on different platforms.

Appendix A: Glossary

TERM / ACRONYM	DEFINITION
ACID	<i>Properties such as Atomicity, Consistency, Isolation, and Durability that guarantee safety of transaction reliability in a database.</i>
JWT(JSON Web Token)	<i>A standard for sharing information between a client and a server for authentication.</i>
PostgreSQL	<i>An open-source relational database management system that is used for data storage.</i>
WCAG 2.1	<i>Guidelines for Web Content Accessibility.</i>
i18n	<i>The process for designing software applications that can be adapted to various languages without any modification.</i>
API	<i>Set of protocols that allow the frontend and the backend server to exchange data.</i>
RESTful API	<i>An architecture for APIs that use HTTP requests to access data.</i>
OpenAPI/Swagger	<i>A framework for describing, producing, consuming, and visualizing RESTful API web services.</i>
KVKK/GDPR	<i>Digital Privacy and Data Protection laws in Turkey and Europe.</i>

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